



KAMBALA

Student Number: _____

AUGUST 2007
YEAR 12
HSC ASSESSMENT TASK#4
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks – 84

- Attempt Questions 1-7.
- All questions are of equal value.

Mathematics Extension 1 HSC Trial

Total marks – 84

Attempt Questions 1-7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

QUESTION 1 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Solve $\frac{4}{x-1} \geq 1$ for x . 3
- (b) Write down the inverse function of $y = e^{2x+1}$. Express your answer as a function of x . 2
- (c) Use the table of standard integrals to evaluate $\int_0^{\frac{\pi}{12}} \sec 3x \tan 3x \, dx$. 2
- (d) Given $f(x) = \tan^{-1}(\sin x)$ find $f'(\pi)$. 2
- (e) The curves $y = \ln x$ and $y = 1 - x^2$ intersect at the point $P(1, 0)$. 3

Find the acute angle between the tangents to the curves at P . Give your answer to the nearest degree.

QUESTION 2 (12 marks) Use a SEPARATE writing booklet.

- (a) The point P divides the interval AB in the ratio $k:1$ where A and B are the points $(4, -1)$ and $(0, 3)$ respectively.

(i) Write down the co-ordinates of P in terms of k .

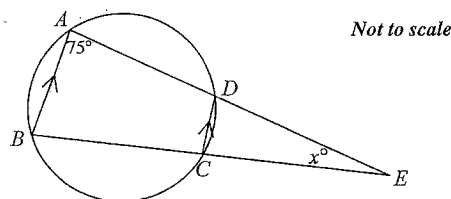
(ii) If P lies on the line $y = 2x$, find k and hence find P .

- (b) The polynomial equation $2x^3 - 4x^2 + 5x - 3 = 0$ has roots α , β and γ .

Find the exact value of $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$.

- (c) Evaluate $\int_0^{\log 3} \frac{e^x}{\sqrt{1+e^x}} dx$, using the substitution $u = e^x$.

- (d) In the diagram below, AB is parallel to DC , $\angle BAD = 75^\circ$ and $\angle DEC = x^\circ$.



Find the value of x , giving reasons for your answer.

QUESTION 3 (12 marks) Use a SEPARATE writing booklet.

- (a) (i) Express $6\cos x + 8\sin x$ in the form $R\cos(x-\alpha)$, where $R > 0$ and α is an acute angle in radians.

- (ii) Hence, or otherwise, solve the equation $6\cos x + 8\sin x = 5$ for $0 \leq x \leq 2\pi$.

- (b) Find $\frac{d}{dx}(\sin^{-1}(x-1))$ in its simplest form, and hence deduce that

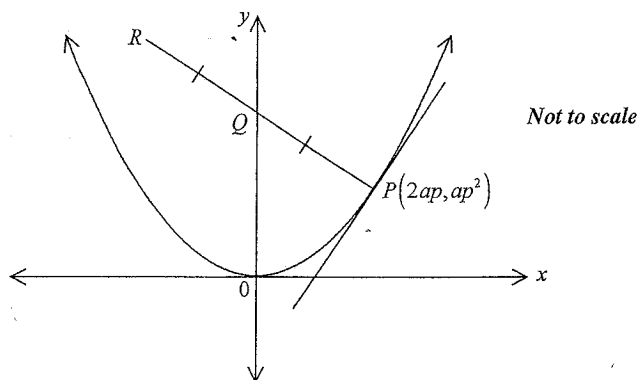
$$\int_{\frac{1}{2}}^1 \frac{dx}{\sqrt{x(2-x)}} = \frac{\pi}{6}$$

- (c) (i) Sketch the graph of $y = \cos x$, $-\pi \leq x \leq \pi$, and use this graph to show that $\cos x + x = 0$ has only one solution.

- (ii) Use Newton's method with a first approximation of $x = -1$ to find a better approximation to the root of $\cos x + x = 0$.

QUESTION 4 (12 marks) Use a SEPARATE writing booklet.

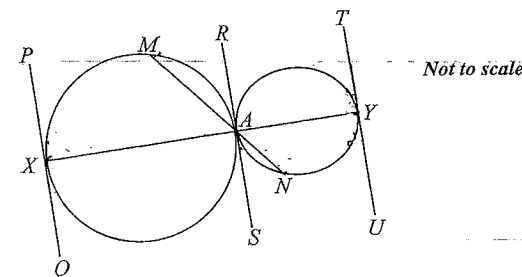
- (a) Using $t = \tan \frac{x}{2}$, prove that $\frac{1 + \cos x}{1 - \cos x} = \cot^2 \frac{x}{2}$ 2
- (b) Use the identity $\sin 2x = 2 \sin x \cos x$ to find $\int \sin^2 x \cos^2 x \, dx$. 3
- (c) The normal to the parabola $x^2 = 4ay$ at the point $P(2ap, ap^2)$ meets the y-axis at Q and is produced to R so that $PQ = QR$ as shown.



- (i) Show that the equation of the normal to the parabola $x^2 = 4ay$ at the point $P(2ap, ap^2)$ is $x + py = 2ap + ap^3$. 2
- (ii) Find the co-ordinates of Q . 1
- (iii) Show that the co-ordinates of R are $(-2ap, 4a + ap^2)$. 2
- (iv) Find the cartesian equation of the locus of R . 2

QUESTION 5 (12 marks) Use a SEPARATE writing booklet.

- (a) A vessel used for water is a solid of revolution formed by the rotation of the parabola $9y = 8x^2$ about the y-axis. The depth of the vessel is 8 cm. 2
- (i) Show that when the height of the water is h cm from its base the volume is given by $V = \frac{9}{16} \pi h^2 \text{ cm}^3$. 2
- (ii) Show that when the vessel is half full the height of the water is $4\sqrt{2}$ cm. 1
- (iii) If water is poured into the vessel at a rate of $20 \text{ cm}^3/\text{sec}$, find the rate at which the level of water is rising when the vessel is half full. 2
- (b) (i) Sketch the graph of $y = \tan x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. 1
- (ii) Hence, or otherwise, find values of x in the domain $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ such that the series $1 + \sqrt{3} \tan x + 3 \tan^2 x + 3\sqrt{3} \tan^3 x + \dots$ has a limiting sum. 2
- (c) Two circles touch at the point A . Lines through A meet the circles at X and Y and at M and N respectively, as shown. RS , the tangent at A is shown. 4



Copy the diagram into your writing booklets.

Prove that PQ and TU , the tangents at X and Y respectively, are parallel.

QUESTION 6 (12 marks) Use a SEPARATE writing booklet. **Marks**

- (a) The rate of increase of a population P of sandflies at Sand Fly Point on the Milford Track is proportional to the difference between the population, P , and 2000. This rate can be expressed by the differential equation $\frac{dP}{dt} = k(P - 2000)$, where k is a constant and t represents time in weeks.
- (i) Show that $P = 2000 + Ae^{kt}$, where A is a constant, satisfies the differential equation. 1
- (ii) Initially, the population is 2500 and two weeks later it had increased to 5000. Find the values of A and k . 2
- (iii) After how many weeks will the population of sandflies exceed 10 000? 1
- (b) A particle moves in a straight line so that its velocity after t seconds is $v \text{ ms}^{-1}$ and its displacement from the origin is x metres.
- (i) Given that $\ddot{x} = 10x - 2x^3$ and that $v = 0$ when $x = -1$, find v^2 in terms of x . 2
- (ii) Explain why the motion cannot exist between $x = -1$ and $x = 1$. 1
- (iii) Describe briefly what would have happened if the motion had commenced at $x = 0$ with $v = 0$. 1
- (c) (i) Use Mathematical Induction to prove that for all positive integers n 3

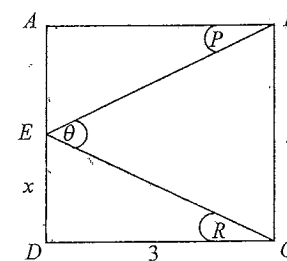
$$4(1^3 + 2^3 + 3^3 + \dots + n^3) = n^2(n+1)^2$$

(ii) Hence find the value of $\lim_{x \rightarrow \infty} \left(\frac{1^3 + 2^3 + 3^3 + \dots + n^3}{n^4} \right)$ 1

QUESTION 7 (12 marks) Use a SEPARATE writing booklet. **Marks**

- (a) (i) Sketch a possible shape of the continuous function $y = f(x)$ given that $f(0) = 3$, $\lim_{x \rightarrow \infty} f(x) = 0$ and that it has only one zero, at $x = -1$. 1
- (ii) Explain why $y = f(x)$ has an inflexion point to the right of $x = -1$. 1
- (b) A particle is moving about the origin O according to the rule $x = 4 \sin 3t$ where x metres is the displacement from O at time t seconds.
- (i) Show that the motion is simple harmonic. 2
- (ii) Determine when the particle is at $x = 2$ metres for the first time. 2
- (iii) Write down the maximum speed of the particle. 1
- (c) A billboard, BC , is 4 metres high and is positioned vertically and parallel to a highway at a height which maximises $\tan \theta$, where θ is the angle subtended at the eyes of the passengers on the top deck of passing double-decker buses. The billboard must be 3 metres from the passengers.

Let $\angle ABE = P$, $\angle DCE = R$ and $ED = x$.



Not to scale

- (i) Copy the diagram into your answer booklet and express θ in terms of P and R . 1
- (ii) Show that $\tan \theta = \frac{12}{9 - 4x + x^2}$. 4

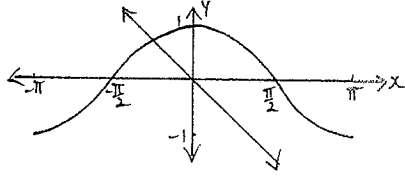
Hence find the value of x which maximises $\tan \theta$.

End of Assessment

Question	Solutions	Marks	Marking Criteria
1 (a)	$\frac{4}{x-1} \geq 1 \quad x \neq 1$ $4(x-1) \geq (x-1)^2$ $(x-1)^2 - 4(x-1) \leq 0$ $(x-1)(x-5) \leq 0$ $(x-1)(x-5) \leq 0$ $1 < x \leq 5$		
(b)	inverse: $x = e^{2y+1}$ $2y+1 = \ln x$ $2y = \ln x - 1$ $y = \frac{1}{2}(\ln x - 1)$		
(c)	$\int_0^{\frac{\pi}{2}} \sec 3x \tan 3x \, dx$ $= \left[\frac{1}{3} \sec 3x \right]_0^{\frac{\pi}{2}}$ $= \frac{1}{3} \sec \frac{\pi}{4} - \frac{1}{3} \sec 0$ $= \frac{1}{3}(\sqrt{2} - 1)$		
(d)	$f(x) = \tan^{-1}(\sin x)$ $f'(x) = \frac{1}{1+\sin^2 x} \times \cos x$ $f'(\pi) = \frac{\cos \pi}{1+\sin^2 \pi}$ $= -1$		
(e)	$y = \ln x \Rightarrow y' = \frac{1}{x}$ $m_T = 1$ $y = 1-x^2 \Rightarrow y' = -2x$ $m_T = -2$ $\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $ $= \left \frac{1 - (-2)}{1 + 1 \times (-2)} \right $ $= 3 \quad \therefore \theta = 72^\circ$		

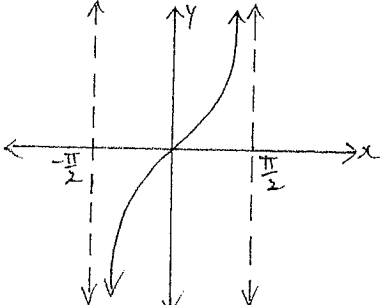
Question	Solutions	Marks	Marking Criteria
2 (a) (i)	$x_p = \frac{4x+0 \times k}{k+1} \quad y_p = \frac{-1 \times 1 + 3 \times k}{k+1}$ $P \left(\frac{4}{k+1}, \frac{3k-1}{k+1} \right)$		
(ii)	P lies on $y = 2x$ $\therefore \frac{3k-1}{k+1} = 2 \times \frac{4}{k+1}$ $3k-1 = 8$ $k = 3$ $\therefore P = (1, 2)$		
(b)	$\frac{1}{\alpha\beta} + \frac{1}{\alpha\delta} + \frac{1}{\beta\delta} = \frac{\delta + \beta + \alpha}{\alpha\beta\delta}$ $= \frac{-(-4)}{2} - \frac{(-3)}{2}$ $= \frac{4}{3}$		
(c)	$\int_0^{\ln 3} \frac{e^x}{\sqrt{1+e^x}} \, dx$ $= \int_1^3 (1+u)^{-\frac{1}{2}} \, du$ $= \left[\frac{(1+u)^{\frac{1}{2}}}{\frac{1}{2}} \right]_1^3$ $= 2(\sqrt{4} - \sqrt{2})$ $= 4 - 2\sqrt{2}$ $u = e^x \quad \frac{du}{dx} = e^x$ $x=0 \quad u=1$ $x=\ln 3 \quad u=3$ $\sqrt{1+e^x} = (1+u)^{\frac{1}{2}}$		
(d)	$\angle EDC = 75^\circ$ corresp. angles $AB \parallel CD$ $\angle ECD = 75^\circ$ ext. angle of cyclic quad equals int. opposite $\angle$ $\therefore x = 180^\circ - (2 \times 75^\circ)$ $\angle$ sum of Δ $= 30^\circ$		

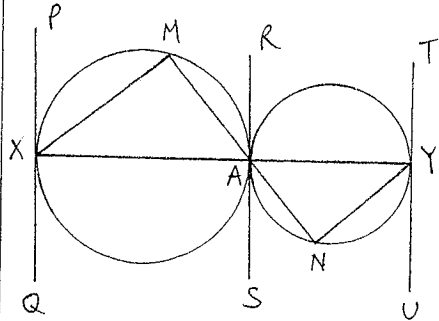
Question	Solutions	Marks	Marking Criteria
3 (a) (i)	$R \cos(x-\alpha) = R \cos x \cos \alpha + R \sin x \sin \alpha$ $6 \cos x + 8 \sin x = R \cos x \cos \alpha + R \sin x \sin \alpha$ $\therefore R \cos \alpha = 6$ $R \sin \alpha = 8$ $\text{square + add} \Rightarrow R = 10 \quad (R > 0)$ $\tan \alpha = \frac{4}{3} \Rightarrow \alpha = \tan^{-1}\left(\frac{4}{3}\right)$ $= 0.927\dots$ $6 \cos x + 8 \sin x = 10 \cos(x - 0.927)$		
(ii)	$6 \cos x + 8 \sin x = 5$ $\text{ie } 10 \cos(x - 0.927) = 5$ $\cos(x - 0.927) = \frac{1}{2}$ $x - 0.927 = \frac{\pi}{3}, \frac{5\pi}{3}$ $x = \frac{\pi}{3} + 0.927, \frac{5\pi}{3} + 0.927$ $x = 1.974, 6.163$		
(b)	$\frac{d}{dx} (\sin^{-1}(x-1)) = \frac{1}{\sqrt{1-(x-1)^2}} \times 1$ $= \frac{1}{\sqrt{1-(1-2x+x^2)}}$ $= \frac{1}{\sqrt{2x-x^2}}$ $= \frac{1}{\sqrt{x(2-x)}}$ $\therefore \int_{\frac{1}{2}}^1 \frac{dx}{\sqrt{x(2-x)}} = \left[\sin^{-1}(x-1) \right]_{\frac{1}{2}}^1$ $= \sin^{-1}(0) - \sin^{-1}\left(-\frac{1}{2}\right)$ $= 0 - \left(-\frac{\pi}{6}\right)$ $= \frac{\pi}{6}$		

Question	Solutions	Marks	Marking Criteria
3 (c) (i)	 $\cos x + x = 0$ ie solve simult. $y = \cos x$ and $y = -x$ only 1 point of intersection $\therefore \cos x + x = 0$ has only one solution		
(ii)	$f(x) = \cos x + x \quad f(-1) = -0.45\dots$ $f'(x) = -\sin x + 1 \quad f'(-1) = 1.84\dots$ $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ $= -1 - \frac{-0.45\dots}{1.84\dots}$ $= -0.75$ $\therefore$ better approximation is -0.75		

Question	Solutions	Marks	Marking Criteria
4 (a)	$t = \tan \frac{x}{2} \quad \cos x = \frac{1-t^2}{1+t^2}$ $\text{LHS} = \frac{1+\cos x}{1-\cos x}$ $= \frac{1+\frac{1-t^2}{1+t^2}}{1-\frac{1-t^2}{1+t^2}}$ $= \frac{\frac{1+t^2+1-t^2}{1+t^2}}{\frac{1+t^2-(1-t^2)}{1+t^2}}$ $= \frac{2}{2t^2}$ $= \frac{1}{t^2}$ $= \frac{1}{\tan^2 \frac{x}{2}}$ $= \cot^2 \frac{x}{2}$		
(b)	$\sin 2x = 2 \sin x \cos x$ $\sin^2 x \cos^2 x = (\sin x \cos x)^2$ $= \left(\frac{1}{2} \sin 2x\right)^2$ $= \frac{1}{4} \sin^2 2x$ $\int \sin^2 x \cos^2 x dx = \frac{1}{4} \int \sin^2 2x dx$ $\sin^2 2x = \frac{1}{2} (1 - \cos 4x)$ $\therefore \int \sin^2 x \cos^2 x dx = \frac{1}{8} \int (1 - \cos 4x) dx$ $= \frac{1}{8} \left(x - \frac{1}{4} \sin 4x\right) + C$		
(c) (i)	$x^2 = 4ay \Rightarrow y = \frac{x^2}{4a}$ $y' = \frac{x}{2a} \quad \therefore m_T = p \quad m_N = -\frac{1}{p}$ eq. normal $y - ap^2 = -\frac{1}{p}(x - 2ap)$ $py - ap^3 = -x + 2ap$ ie $x + py = 2ap + ap^3$		

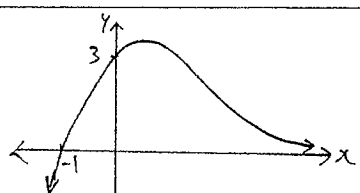
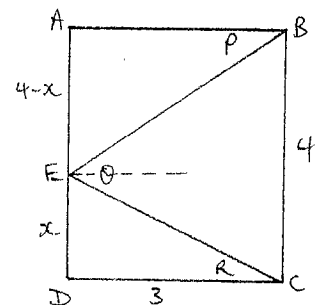
Question	Solutions	Marks	Marking Criteria
4 (c) (ii)	at Q (on the normal) $x = 0$ $\therefore py = 2ap + ap^3$ $y = 2a + ap^2$ Q (0, $2a + ap^2$)		
(iii)	Q is midpt. of RP $\therefore 0 = \frac{x_R + 2ap}{2} \Rightarrow x_R = -2ap$ $2a + ap^2 = \frac{y_R + ap^2}{2} \Rightarrow y_R = 4a + ap^2$ $\therefore R(-2ap, 4a + ap^2)$		
(iv)	$x = -2ap \Rightarrow p = -\frac{x}{2a}$ $y = 4a + ap^2$ $y = 4a + a\left(-\frac{x}{2a}\right)^2$ $y = 4a + \frac{x^2}{4a}$ $4ay = 16a^2 + x^2$ $x^2 = 4ay - 16a^2$ $x^2 = 4a(y - 4a)$		

Question	Solutions	Marks	Marking Criteria
5(a)(i)	$V = \pi \int_0^h x^2 dy \quad \text{volume about y-axis}$ $= \pi \int_0^h \frac{9y}{8} dy$ $= \pi \left[\frac{9y^2}{16} \right]_0^h$ $= \frac{9}{16} \pi h^2 \text{ cm}^3$		
(ii)	vessel is full when $h=8$ $V = \frac{9}{16} \pi \times 8^2 = 36\pi$ $\therefore \frac{1}{2} V = 18\pi$ $18\pi = \frac{9}{16} \pi h^2$ $h^2 = 32$ $h = \sqrt{32} = 4\sqrt{2} \text{ cm}$		
(iii)	$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt} \quad \text{find } \frac{dh}{dt} \text{ when } h=4\sqrt{2}$ $\frac{dV}{dt} = 20 \text{ cm}^3/\text{s}$ $\frac{dV}{dh} = \frac{9}{8} \pi h$ $20 = \frac{9}{8} \pi \times 4\sqrt{2} \times \frac{dh}{dt}$ $\frac{dh}{dt} = \frac{40}{9\pi\sqrt{2}} = \frac{20\sqrt{2}}{9\pi} \text{ cm/s}$		
(b)(i)			

Question	Solutions	Marks	Marking Criteria
5(b)(ii)	$1 + \sqrt{3} \tan x + 3 \tan^2 x + \dots$ For a limiting sum $-1 < r < 1$ $\therefore -1 < \sqrt{3} \tan x < 1$ $-\frac{1}{\sqrt{3}} < \tan x < \frac{1}{\sqrt{3}}$ $-\frac{\pi}{6} < x < \frac{\pi}{6}$		
(c)	 Join XM and YN $\angle PXM = \angle MAX \quad \text{L between tangent + chord} = \text{L in the alternate segment}$ $\angle MAX = \angle YAN \quad \text{vertically opposite}$ $\angle YAN = \angle UYN \quad \text{alt. segment theory}$ $\therefore \angle PXM = \angle UYN$ $\angle SAN = \angle NYA \quad \text{alt. segment theory}$ $\angle RAM = \angle SAN \quad \text{vert. opposite}$ $\angle RAM = \angle MXA \quad \text{alt. segment theory}$ $\therefore \angle MXA = \angle NYA$ $\angle PXA = \angle PXM + \angle MXA$ $\angle UYA = \angle UYN + \angle NYA$ $\therefore \angle PXA = \angle UYA$ alternate angles are equal $\therefore PQ \parallel TU$		When circles touch the line of centres passes through the point of contact. $\therefore AX$ and AY are diameters $\therefore \angle PXA = 90^\circ$ and $\angle UYA = 90^\circ$ tangent to a circle is perpendicular to the radius drawn to the point of contact. $\angle PXA = \angle UYA$ alternate angles are equal $\therefore PQ \parallel TU$

Question	Solutions	Marks	Marking Criteria
6 (a) (i)	$P = 2000 + Ae^{kt} \Rightarrow Ae^{kt} = P - 2000$ $\frac{dP}{dt} = kAe^{kt}$ $= k(P - 2000)$		
(ii)	$t = 0 \quad P = 2500$ $2500 = 2000 + Ae^0$ $A = 500$ $t = 2 \quad P = 5000$ $5000 = 2000 + 500e^{2k}$ $e^{2k} = 6$ $2k = \ln 6$ $k = \frac{1}{2} \ln 6 = 0.895 \dots$		
(iii)	$P = 10000$ $10000 = 2000 + 500e^{\frac{1}{2} \ln 6 t}$ $e^{\frac{1}{2} \ln 6 t} = 16$ $\frac{1}{2} \ln 6 t = \ln 16$ $t = \frac{2 \ln 16}{\ln 6}$ $= 3.09 \text{ weeks}$ in the 3rd week		
(b) (i)	$\ddot{x} = 10x - 2x^3$ $\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 10x - 2x^3$ $\frac{1}{2} v^2 = 5x^2 - \frac{x^4}{2} + C$ $v = 0 \quad x = -1 \Rightarrow 0 = 5 - \frac{1}{2} + C$ $C = -4\frac{1}{2}$ $\frac{1}{2} v^2 = 5x^2 - \frac{x^4}{2} - \frac{9}{2}$ $v^2 = 10x^2 - x^4 - 9$		
(ii)	For $-1 < x < 1 \quad v^2 < 0$ which is possible $\therefore$ motion doesn't exist between $x = -1$ and $x = 1$		

Question	Solutions	Marks	Marking Criteria
6 (b) (ii)	Solve $v^2 < 0$ $10x^2 - x^4 - 9 < 0$ $\Leftrightarrow x^4 - 10x^2 + 9 > 0$ $(x^2 - 9)(x^2 - 1) > 0$ $x < -3, -1 < x < 1, x > 3$		
(iii)	$\frac{1}{2} v^2 = 5x^2 - \frac{x^4}{2} + C$ if $x = 0$ and $v = 0$ then $\ddot{x} = 0$ stationary particle <u>and</u> no acceleration $\therefore$ particle would not move.		
(c) (i)	Show true for $n = 1$ LHS = 4 RHS = $1^2(2)^2 = 4$ LHS = RHS $\therefore$ true for $n = 1$ Assume true for $n = k$ $\Leftrightarrow 4(1^3 + 2^3 + \dots + k^3) = k^2(k+1)^2$ Show true for $n = k+1$ $\Leftrightarrow 4(1^3 + 2^3 + \dots + k^3 + (k+1)^3) = (k+1)^2(k+2)^2$ LHS = $4(1^3 + 2^3 + \dots + k^3) + 4(k+1)^3$ $= k^2(k+1)^2 + 4(k+1)^3$ $= (k+1)^2(k^2 + 4k + 4)$ $= (k+1)^2(k+2)^2$ $= \text{RHS}$ $\therefore$ true for $n = k+1$ if true for $n = k$ since true for $n = 1$ also true for $n = 1+1 = 2$ $\therefore$ thus true for $n = 2+1 = 3$ and so on for all positive integral n .		
(ii)	$\frac{1^3 + 2^3 + \dots + n^3}{n^4} = \frac{\frac{1}{4}n^2(n+1)^2}{n^4} = \frac{\frac{1}{4}(n+1)^2}{n^2}$ $\lim_{n \rightarrow \infty} \frac{\frac{1}{4}(n+1)^2}{n^2} = \lim_{n \rightarrow \infty} \frac{\frac{1}{4}(n^2 + 2n + 1)}{n^2}$ $= \lim_{n \rightarrow \infty} \frac{1}{4} \left(1 + \frac{2}{n} + \frac{1}{n^2} \right)$ $= \frac{1}{4}$ since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ $\therefore \lim_{n \rightarrow \infty} \frac{1^3 + 2^3 + \dots + n^3}{n^4} = \frac{1}{4}$		

Question	Solutions	Marks	Marking Criteria
7 (a) (i)			
(ii)	concave down around $x = -1$ passes through $(0, 3)$ but to return to x-axis ($f(x) \rightarrow 0$ from above) must change from concave down to concave up ie pt. of inflexion to the right of $x = -1$		
(b) (i)	$x = 4 \sin 3t$ $\dot{x} = 12 \cos 3t$ $\ddot{x} = -36 \sin 3t$ $= -9(4 \sin 3t)$ $= -9x$ which is of the form $\ddot{x} = -n^2 x$ $\therefore$ motion is simple harmonic		
(ii)	$2 = 4 \sin 3t$ $\sin 3t = \frac{1}{2}$ $3t = \frac{\pi}{6}$ $t = \frac{\pi}{18} \text{ s}$		
(iii)	max. speed 12 m/s		
(c) (i)	 $\theta = P + R$		

Question	Solutions	Marks	Marking Criteria								
7 (c) (ii)	$\tan \theta = \tan (P + R)$ $= \frac{\tan P + \tan R}{1 - \tan P \tan R}$ $= \frac{\frac{4-x}{3} + \frac{x}{3}}{1 - \frac{4-x}{3} \times \frac{x}{3}}$ $= \frac{\frac{4}{3}}{\frac{9 - (4x - x^2)}{9}}$ $= \frac{12}{9 - 4x + x^2}$ differentiate $\tan \theta$ wrt x $\frac{d(\tan \theta)}{dx} = -12(9 - 4x + x^2)^{-2} \times (-4 + 2x)$ $= \frac{48 - 24x}{(9 - 4x + x^2)^2}$ max when $\frac{d(\tan \theta)}{dx} = 0$ ie $48 - 24x = 0$ $x = 2$ test <table border="1" data-bbox="1433 957 1747 1101"> <tr> <td>x</td> <td>2^-</td> <td>2</td> <td>2^+</td> </tr> <tr> <td>$\frac{d(\tan \theta)}{dx}$</td> <td>$+$</td> <td>$0$</td> <td>$-$</td> </tr> </table> / max \ $\tan \theta$ is a maximum when $x = 2$	x	2^-	2	2^+	$\frac{d(\tan \theta)}{dx}$	$+$	0	$-$		
x	2^-	2	2^+								
$\frac{d(\tan \theta)}{dx}$	$+$	0	$-$								